

COMP27112 Introduction to Visual Computing

Laboratory 4

1 - Canny has two thresholds that control the edge thresholding process. What is their purpose?

The 2 parameters are for finding out which edges are more likely (strong) or less likely (weak) to be an edge.

- Pixels with a gradient above the second (200) threshold are strong edges.
- Pixels with a gradient below the first (100) threshold are non-edges.
- Pixels with a gradient in between the two threshold values are weak edges (100-200)

With this information, the algorithm will connect the weak edges to the strong edges to make the contours of the image.

2 - What is the purpose of the aperture parameter? What is the result of changing it from 3 to 5, 7, 9 or greater?

The aperture parameter sets the size of the Sobel kernel to compute the image gradient by calculating the first derivative. This value is used for determining the level of detail. A high value can find more fine details (but could increase noise due to false positives). A low value can make it smoother to detect more robust contours (but therefore may miss key details) For example (experimenting on horizon3.jpg and the binary value of 159):

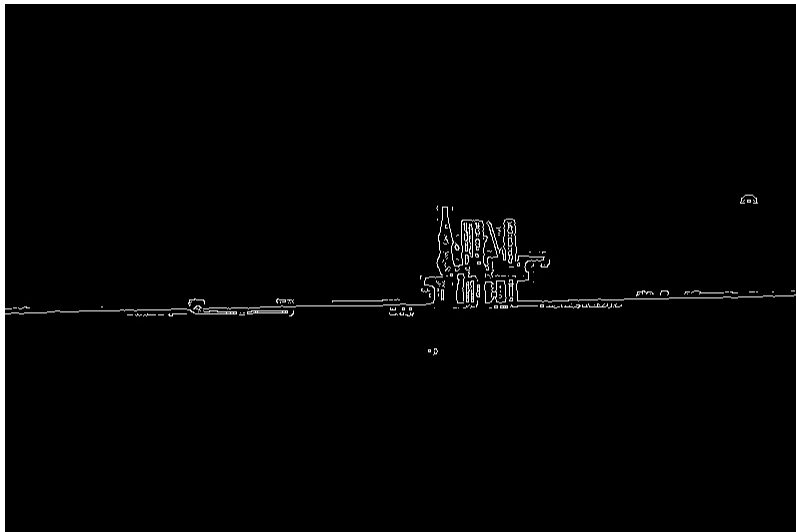
- When setting the aperture parameter to 3, the edges seem quite clean and capture the horizon well in an almost straight line there is not much noise (there is a pixel on the top right which I will continue to refer to as an example of the noise which appears small for now).



- When setting the aperture parameter to 5, the edges appear to be less clean, the pixel on the top right of the image gets slightly bigger and the final line for the horizon image is not created.



- When setting the aperture parameter to 7, the edges appear quite thick with noise and clutter, the centre image of the oil rig also has a blur to it caused by the increased number of noisy edges. Furthermore, the pixel on the top right of the image also appears much larger. However, the final line for the horizon image is created.



- When setting the aperture parameter to 9(+), I get an error: 'Aperture size should be odd between 3 and 7 in function 'Canny'".

3 - The Hough transform has two parameters that specify the resolution of the accumulator. Their default values are 1 and $\pi/180$. What is the effect of increasing the first and reducing the second?

Original image for comparison to changes below:



For example (experimenting on horzion3.jpg and the binary value of 159):

Increasing 1 to 50:

- Increasing rho will (radial distance resolution) will reduce the resolution of the accumulator in the radial direction causing less precise line detection and merging close lines.
- This makes the algorithm more robust against noise but may miss nearby lines.

Decreasing CV_PI/180 to CV/2880:

- Decreasing theta increases the precision of line orientation detection and separation of similar lines.
- Increases accuracy but can be more affected by noise.



4 - The Hough transform has a pair of parameters that determine the minimum length of a line that can be accepted, and the maximum gap between two segments if they are to be considered part of the same line. What is the effect of changing these values?

Original image for comparison to changes below:



minLen is the minimum line length and maxGap is the maximum allowed gap between line segments.

For example (experimenting on horzion3.jpg and the binary value of 159):

Increasing minLen from 50 to 100:

- Only lines greater than a length of 100 are shown. This decreases the number of lines which could make the program miss key data.



Decreasing minLen from 50 to 1:

- All lines greater than a length of 1 show, which increases the number of lines in the image. However, this could lead to an increase in noise.



Increasing maxGap from 10 to 20:

- This allows for more lines to fit next to each other but can increase noise resulting in a worse result.



decreasing maxGap from 10 to 1:

- This means that there is less possible space allowed between two lines. In this example, there are no lines shown. In another scenario for more fine detailing needed, this could be useful.



5 - How close are the computed horizons to where you think the horizon should be?
What might cause any discrepancy?

The computed horizons are very close to where the horizon should be. However, there is some gap in the horizon1.jpg and horizon2.png samples. This is because as I have explained in my previous answers, the parameters which fit one image perfectly, may not fit other images as well (this can lead to overfitting). Adjusting the parameters above could help fine-tune this however, I think the parameters I used are best suited for the average result of all three of these given images.

